

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING (5 Questions)

Course Code:	ITA0502	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation(BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	20% of CIA	Total Marks:	50 (5 Questions × 10 Marks)
CO2 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images..(BL3)		
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Section / Batch:	810t c/Hostel/2024	Date:	

Code of Conduct
 I, Y. Thanushma/192425355 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y. Thanushma

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Grey Level Transformations	Enhancing image brightness and contrast using negative, log, power-law (gamma), and contrast stretching transformations.	1	10
Histogram Processing Techniques	Improving image contrast through histogram analysis, histogram equalization, and histogram matching.	2	10
Spatial Domain Enhancement	Reducing noise and enhancing image quality using smoothing filters and sharpening filters.	3	10
Frequency Domain Enhancement	Enhancing images by applying low-pass and high-pass filters.	4	10
Image Segmentation	Separating objects from the background using point, line, and edge detection, thresholding, region-based segmentation, edge linking, and boundary detection techniques.	5	10
GRAND TOTAL:			50 Marks

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Grey Level Transformation

A grayscale image is enhanced using a negative transformation.

(a) Explain the purpose of grey level transformation in image enhancement

(b) Apply the negative transformation:

$$s = 255 - r$$

to the following pixel values:

$$r = 50, 100, 150, 200$$

Compute the transformed pixel values and interpret the effect on the image.

2. Histogram Equalization

A 4-bit grayscale image has the following histogram:

Intensity	Frequency
0	4
1	6
2	10
3	6

- (a) Explain why histogram equalization is used for image enhancement.
- (b) Compute the cumulative distribution function (CDF).
- (c) Determine the equalized intensity values and interpret the improvement in image contrast.

3. Gaussian Smoothing

A Gaussian filter is applied to reduce image noise.

(a) Explain why Gaussian filtering is preferred over averaging filtering.

(b) Apply the Gaussian kernel

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

to the image region

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Compute the new center pixel value and interpret the smoothing effect.

4. Edge Detection Using Sobel Operator

An image is processed using the horizontal Sobel operator.

(a) Explain the concept and purpose of edge detection.

(b) Apply the horizontal Sobel mask

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

to the image

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Compute the gradient at the center and interpret the result.

5. Contrast Stretching

A grayscale image has intensity values in the range 50–150.

(a) Explain the need for contrast stretching in image enhancement.

(b) Using linear contrast stretching to map the range [50,150] to [0,255], compute the transformed values for

$$r = 50, 100, 150$$

Show the calculations and interpret the improvement in image contrast.

Rubrics for Calculation

							Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem		
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method		
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process		
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations		
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Name

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MCS-1

Y. Thanushma

Course Name

Computer Vision

Matrion Number - M2425355

Course Faculty

Dr. Jini C. S.

Course Code - ITAD503

CO-2 Assessment

1) Gray level Transformation

a) Purpose of gray level transformation image enhancement.

Gray level transformation is used to modify the intensity values (brightness) of pixels in a gray scale image to improve visual appearance. It helps in :-

- * Enhancing image contrast
- * Highlighting important details
- * Improving visibility of objects
- * Making images easier to analyze.

b) Negative Transformation

Formula: $S = 255 - r$

where :- r = original pixel value
 S = Transformed pixel value

Original Pixel value (r)	Transformed pixel value ($S = 255 - r$)
50	205
100	155
150	105
200	55

Interpretation :-

The negative transformation inverts the grayscale image:

- * Dark pixels become bright.
- * Bright pixels become dark
- * light areas appear dark, and dark areas appear light.

This transformation is useful for enhancing details in images where dark regions need to be emphasized, such as medical or X-ray images.

② Histogram Equalization

a) Why it is used:-

- * Improves contrast of an image
- * Spread pixel intensity values over available range
- * Enhance details in dark and bright regions
- * Make the image easier to interpret.

b) Compute CDF

$$\text{CDF} = \text{CDF}(i-1) + \text{Frequency}(i)$$

Intensity	Frequency	Cumulative Frequency (CDF)
0	4	4
1	6	$4+6 = 10$
2	10	$10+10 = 20$
3	6	$20+6 = 26$

Total number of pixels (N) = $4+6+10+6 = 26$

Equalized Intensity values.
 For a 4-bit gray scale image, the number of intensity values is:

$$L = 2^4 = 16$$

Formula:-
$$S_k = \frac{CDF_k}{N} \times (L-1)$$

$$N = 26$$

$$L-1 = 15$$

PDF?

Intensity (r)	CDF	Equalized value $S = \frac{CDF}{26} \times 15$	Rounded value
0	4	2.31	2
1	10	5.77	6
2	20	11.54	12
3	26	15.00	15

Interpretation:-

* The original intensity values (0-3) are spread over a wider range (2-15)

* This increases the image contrast

* Dark and bright regions become more distinguishable

* Fine details become more visible, resulting in a clearer image.

③ Gaussian Smoothing

a) Why is gaussian filtering preferred over Averaging Filtering

* It gives more weight to the center pixel and less weight to the neighbouring pixels.

* It reduces noise while preserving image edges better than averaging filter.

* It produces a smoother and more natural-looking image.

b) Applying the Gaussian kernel.

Gaussian kernel:-

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image region:-

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Multiplying corresponding elements:-

$$(1 \times 10) + (2 \times 20) + (1 \times 10) + (2 \times 20) + (4 \times 40) + (2 \times 20) + (1 \times 10) + (2 \times 20) + (1 \times 10)$$

$$= 10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10 = 360$$

$$\text{Divide by } 16 = \frac{360}{16} = 22.5$$

Rounded value :- 23

New center pixel value = 23

Interpretation :-

- * The original center pixel value (40) is smoothed to 23
- * Noise is reduced by averaging neighbouring pixels with higher weight given to the center.
- * The image becomes smoother while preserving edges better than a simple averaging filter.

④ Edge detection using Sobel Operator

a) Concept and purpose of edge detection

Edge detection is a technique used to identify boundaries (edges) in an image where there is a sudden change in pixel intensity.

Purpose:-

- * Detect object boundaries.

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Explain important features

- * simplify image analysis
- * used in image segmentation, object detection and computer vision.

b) Apply horizontal Sobel mask

Horizontal Sobel mask:-

Image:-

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Multiply corresponding elements:-

$$(-1 \times 10) + (-2 \times 20) + (-1 \times 30) + (0 \times 20) + (0 \times 20) + (0 \times 20) + (1 \times 30) + (2 \times 30) + (1 \times 30)$$

$$= -10 - 20 - 10 + 0 + 0 + 0 + 30 + 60 + 30$$

$$= 80$$

Gradient at the Centre:- $G_x = 80$

Interpretation :-

* High gradient value (80) indicates a strong horizontal edge.

* The pixel intensity changes from 10 $\rightarrow$ 20 $\rightarrow$ 30, so the Sobel operator detects a clear edge.

* This means there is a significant intensity transition in the image, indicating the presence of an edge.

c) Contrast Stretching:-

Need for Contrast Stretching:-

Contrast stretching is an image enhancement technique used to:

- Impairment contrast at an average
- Expand a narrow range of intensity values to the full intensity range
- Make dark regions darker and bright regions brighter
- Enhance the visibility of image details

(b) Linear Contrast Stretching

Formula:

$$s = \frac{(r - r_{min})}{(r_{max} - r_{min})} \times 255$$

Example:

$$r_{min} = 50$$

$$r_{max} = 180$$

For

$$s = \frac{(r - 50)}{(180 - 50)} \times 255$$

Example: Calculate for $r = 50$

$$s = \frac{50 - 50}{180 - 50} \times 255 = 0$$

$$r = 100 \quad s = \frac{100 - 50}{180 - 50} \times 255 = 97.5$$

$$r = 180 \quad s = \frac{180 - 50}{180 - 50} \times 255 = 255$$

and transformed values to
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4) Interpretation:-

- * The original intensity range 50-150 is stretched to the full range 0-255
- * This image increases the image contrast
- * Darker, bright pixels become brighter and image details become clearer
- * The enhanced image has improved visibility and better distinction with objects between